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EFFECT OF STRESS ON MEMORY

A Review of Research on Memory Formation, Consolidation and Retrieval

Subject: Cognitive Psychology

Project: Review of Research Papers in a Selected Field of Cognitive Psychology

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1. Introduction

Stress is a psychological and physiological response that occurs when an individual faces challenging, demanding or threatening situations. It can influence several cognitive processes, including attention, concentration, learning and memory. In cognitive psychology, memory refers to the process through which information is encoded, stored and retrieved when required.

The relationship between stress and memory has been widely studied because memory plays an important role in everyday functioning. Activities such as remembering information for examinations, learning new skills and recalling past experiences depend on effective memory processes.

When a person experiences stress, the body activates its stress response system and releases hormones such as cortisol and adrenaline. These hormones influence different brain regions, particularly the hippocampus, amygdala and prefrontal cortex, which are associated with memory and other cognitive functions.

Stress does not always have the same effect on memory. Moderate stress may sometimes improve attention and help individuals remember emotionally significant information. However, excessive or prolonged stress can interfere with learning, memory retention and the ability to retrieve previously learned information. For example, a student may prepare well for an examination but struggle to recall answers because of examination anxiety.

Research also suggests that the timing of stress is important. Stress experienced before learning may have different effects from stress experienced after learning or immediately before recalling information. Therefore, understanding stress requires examining its influence on different stages of memory.

The present review examines research on the effects of stress on memory formation, consolidation and retrieval. It also explores the role of stress hormones, compares findings from previous studies and identifies limitations and possible directions for future research.

Objectives of the review

- To understand the relationship between stress and memory.
- To examine how acute and prolonged stress influence memory performance.
- To understand the role of cortisol and brain functioning in memory.
- To compare and integrate findings from previous research studies.
- To identify methodological limitations and suggest areas for future research.

2. Theoretical Background

Several psychological theories and concepts help explain how stress influences memory.

2.1 Yerkes-Dodson Law

The Yerkes-Dodson Law, proposed by Yerkes and Dodson (1908), explains the relationship between arousal and performance. According to this principle, performance may improve as

arousal increases, but only up to an optimal level. Beyond this point, excessive arousal may negatively affect performance.

In relation to memory, a moderate level of stress may increase alertness and attention, helping individuals concentrate on information. However, excessive stress may interfere with concentration and make remembering information difficult. The relationship also depends on the complexity of the task and individual differences.

2.2 Transactional Model of Stress

Lazarus and Folkman (1984) proposed the transactional model of stress, which explains that stress depends partly on how individuals evaluate and interpret a situation.

According to this model, individuals experience stress when they perceive that the demands of a situation exceed their ability to cope with it.

For example, two students may face the same examination, but one may consider it manageable while the other may perceive it as threatening. Their stress levels and memory performance may therefore differ.

This model helps explain why the same stressful situation may affect different individuals in different ways.

2.3 Stress Hormones and Memory

Stress activates the hypothalamic-pituitary-adrenal (HPA) axis, which leads to the release of cortisol. Cortisol is an important stress hormone that influences brain regions involved in memory functioning.

The three important brain regions are:

- Hippocampus: Plays an important role in learning and forming new memories.
- Amygdala: Processes emotional information and influences emotional memories.
- Prefrontal cortex: Supports working memory, attention, planning and decision-making.

Changes in stress hormone levels can influence the functioning of these brain regions. High or poorly timed cortisol responses may interfere with certain memory processes, although the effects depend on the timing and duration of stress.

3. Review of Selected Research Studies

The following six empirical studies and one meta-analysis were selected to understand the different effects of stress on memory. They examine declarative memory, working memory, memory retrieval and memory consolidation.

3.1 Kirschbaum et al. (1996)

Kirschbaum and colleagues conducted two experiments to examine the relationship between cortisol levels and memory performance in healthy adults.

In the first experiment, 13 participants were exposed to a psychosocial stress situation, after which their declarative memory performance was assessed. The researchers observed that participants who showed higher cortisol responses performed more poorly on the memory task.

In the second experiment, 40 healthy participants received either cortisol or a placebo. Their memory performance was assessed approximately one hour later. Participants who received cortisol showed poorer performance in declarative memory and spatial thinking, while procedural memory was not significantly affected.

The findings suggested that increased cortisol levels may negatively influence certain types of memory. However, the study also demonstrated that different memory systems may respond differently to stress.

One limitation was the relatively small sample in the first experiment, which makes it difficult to generalise the findings to a wider population. PubMed +1

3.2 Lupien et al. (1997)

Lupien and colleagues investigated the effects of psychosocial stress on declarative and non-declarative memory among healthy elderly individuals.

The study included 14 participants who were exposed to two conditions: a stressful public-speaking task and a non-stressful attentional task. Researchers measured their memory performance and salivary cortisol levels before and after the tasks.

The results showed that the stressful condition reduced declarative memory performance, whereas non-declarative memory remained relatively unaffected. The researchers also observed differences in cortisol responses among participants.

The study suggested that stress may particularly affect memory processes associated with the hippocampus. It also highlighted the importance of individual differences in stress responses.

However, the sample size was very small and consisted only of elderly participants. Therefore, the findings may not be directly applicable to younger adults or university students. PubMed +1

3.3 Oei et al. (2006)

Oei and colleagues examined the influence of psychosocial stress on working memory and memory retrieval.

The researchers used the Trier Social Stress Test to create a stressful situation. Participants completed working-memory tasks involving different levels of cognitive load, along with declarative memory assessments.

The findings indicated that stress negatively affected working memory, particularly when participants were required to process a high amount of information simultaneously. The study also identified an association between cortisol responses and memory performance.

These findings suggest that stress can make it difficult to manage and process information when cognitive demands are high.

The study is particularly relevant to academic situations because students often need to process, organise and recall large amounts of information under pressure.

However, the laboratory-based nature of the study may not fully represent the complexity of stress experienced in everyday academic situations. PubMed +1

3.4 Wolf et al. (2002)

Wolf and colleagues investigated whether moderate psychosocial stress affects the recall of information learned several weeks earlier.

The study involved 40 participants who learned a list of words either before or after exposure to a laboratory stress situation. Their delayed memory performance was assessed four weeks later.

The researchers found that stress did not significantly affect initial learning or delayed recall. They also found no clear evidence that learning and recalling information under similar stress conditions improved memory performance.

This study is important because its findings differed from some earlier research suggesting that stress consistently impairs memory.

The authors discussed possible explanations, including the moderate cortisol response and the long delay between learning and recall. The findings suggest that previously established memories may not always be affected by later stressful experiences.

A limitation was the relatively small sample size, which may have reduced the ability to detect subtle effects of stress. PubMed +1

3.5 Smeets et al. (2007)

Smeets and colleagues examined whether acute stress could improve memory when the stressful situation and the information being learned were related to each other.

The study included 52 healthy undergraduate students who were divided into three groups: a personality-stress group, a memory-stress group and a no-stress control group.

Participants learned personality-related and memory-related words while being exposed to stressful or control conditions. Their memory was tested through a free-recall task after 24 hours.

The results showed that stress improved the recall of information that was related to the stressful situation, particularly personality-related words in the personality-stress condition.

This study suggests that stress may sometimes strengthen memory when the information being learned is directly connected to the stressful experience. It provides an explanation for why emotionally significant experiences may be remembered more clearly.

However, the study used specific word-learning tasks, so its findings may not apply equally to all types of learning or real-life stressful experiences. PubMed +1

3.6 Smeets (2011)

Smeets investigated whether acute stress affects memory retrieval and whether the effect changes according to the time of day.

The study included 76 participants who were exposed to either a stressful condition or a control condition before a memory test. Participants had learned neutral and negative words 24 hours earlier.

The researchers assessed their ability to recall the previously learned words and also measured cortisol responses.

The findings showed that participants exposed to stress experienced difficulty recalling both neutral and negative words. The negative effect was greater for emotionally negative information. The results also indicated that the time of day did not significantly change the stress-related retrieval impairment.

This study is important because it demonstrates that stress can interfere with the retrieval of previously learned information, even when the information was learned a day earlier.

However, the study focused on word recall, which represents only one aspect of memory. Other forms of memory may respond differently to stress. PubMed +1

3.7 Shields et al. (2017): Meta-analysis

Shields and colleagues conducted a meta-analysis and integrative review to examine the effects of acute stress on episodic memory.

The researchers analysed 113 independent studies involving 6,216 human participants. They examined the effects of stress during memory encoding, after learning, during retrieval and after memory reactivation.

The findings demonstrated that stress occurring before or during learning generally impaired memory, although it could improve memory when the information was directly related to the stressful situation.

Stress occurring shortly after learning could improve memory consolidation under certain conditions. In contrast, stress experienced immediately before or during retrieval generally impaired memory performance.

This research is particularly valuable because it brings together findings from a large number of studies and explains why different experiments have sometimes reported contradictory results.

One limitation is that the studies included in the meta-analysis used different experimental procedures, stress conditions and memory measures. Nevertheless, it provides strong evidence that the timing and context of stress are important factors in understanding memory performance. PubMed +1

4. Comparison and Integration of Research Findings

The reviewed studies demonstrate that stress has a complex relationship with memory. Its effects depend on the type of memory, the intensity of stress and the stage at which stress occurs.

A common finding in the studies conducted by Kirschbaum et al. (1996), Lupien et al. (1997) and Oei et al. (2006) is that stress can negatively influence declarative and working memory. These studies suggest that increased stress and cortisol responses may interfere with the ability to process and remember information.

Similarly, Smeets (2011) demonstrated that acute stress can affect memory retrieval. This finding is particularly relevant to examination situations, where students may experience difficulty recalling information despite having studied it earlier.

However, the findings of Wolf et al. (2002) were different. The researchers did not observe significant impairment in delayed recall following moderate psychosocial stress. This indicates that the negative effects of stress may depend on its intensity, timing and the strength of previously formed memories.

Smeets et al. (2007) further demonstrated that stress may improve memory when the information being learned is related to the stressful situation. Shields et al. (2017) supported this explanation by showing that stress may have different effects during encoding, consolidation and retrieval.

Overall, the studies indicate that stress should not be considered entirely harmful to memory. Instead, its influence varies according to the circumstances in which it occurs. While excessive stress may interfere with learning and recall, certain stressful experiences may strengthen the memory of relevant information.

5. Critical and Methodological Analysis

The selected studies provide useful information about the relationship between stress and memory. One major strength is that several researchers used experimental methods to study stress under controlled conditions. For example, Kirschbaum et al. (1996) and Lupien et al. (1997) measured cortisol levels along with memory performance. This helped researchers understand the relationship between physiological stress responses and cognitive functioning.

Another strength is the use of different memory assessments. The studies examined declarative memory, working memory, episodic memory and delayed recall. This provides a broader understanding of how stress influences different aspects of memory.

However, some methodological limitations can also be identified.

1. Small sample sizes: Some studies, such as Lupien et al. (1997), included very few participants. This limits the generalisability of their findings.

2. Limited participant diversity: Some studies focused on specific age groups or selected healthy individuals. Their findings may not represent the experiences of people from different age groups and backgrounds.

3. Laboratory conditions: Many researchers used artificial stress situations, such as public speaking and laboratory tasks. These may not fully represent the stress experienced in everyday life.

4\. Individual differences: People respond differently to stressful situations. Personality, coping strategies, previous experiences and emotional responses may influence memory performance.

5. Differences in research methods: The studies used different stress-induction techniques, memory tests and testing intervals. This makes direct comparisons between some findings difficult.

6. Limited research on prolonged stress: Many studies focused on short-term stress rather than long-term stress experienced over weeks or months.

Therefore, although the existing research provides important evidence, further studies involving diverse populations and real-life situations are needed.

6. Research Gap and Future Directions

Although several researchers have examined the relationship between stress and memory, certain areas still require further investigation.

One important research gap concerns the effect of academic stress on memory among university students. Many previous studies have examined short-term stress under laboratory conditions. However, less attention has been given to how prolonged academic pressure influences memory formation, retention and retrieval in everyday student life.

University students regularly experience academic pressure due to examinations, assignments, deadlines and expectations regarding their academic performance. Such situations may influence their ability to concentrate, learn and recall information.

Future research could investigate the relationship between academic stress and memory performance among undergraduate and postgraduate students.

Suggested research question:

How does academic stress influence memory formation and retrieval among university students during examination periods?

A possible future study could include students from different academic backgrounds and compare their perceived stress levels with their performance on standardised memory tests.

Researchers could use the Perceived Stress Scale (PSS) to measure perceived stress and suitable memory assessment tasks to examine recall and working memory.

The study could compare memory performance during regular academic periods and examination periods. It could also examine whether students with different coping strategies show differences in memory performance.

Such research would help develop a better understanding of academic stress and its influence on cognitive functioning. The findings may also help educational institutions develop suitable strategies to support students during examination periods.

7. **Conclusion**

The present review examined the relationship between stress and memory through different research studies. The findings indicate that stress has a significant but complex influence on memory functioning.

Several studies suggest that excessive stress and increased cortisol responses may negatively affect working memory, declarative memory and memory retrieval. Stress can make it difficult for individuals to concentrate, process information and recall previously learned material.

However, research also demonstrates that stress does not always produce negative effects. Under certain conditions, particularly when stress occurs shortly after learning, it may improve memory consolidation. The relevance of the stressful experience to the information being learned also plays an important role.

The reviewed studies highlight that the effects of stress depend on its intensity, duration, timing and individual differences. Therefore, it is important to understand stress in relation to different stages of memory rather than considering it entirely harmful.

Further research focusing on academic stress among university students may provide valuable insights into the relationship between stress, learning and memory performance.

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